

COMP 421 - Database Systems

Project 3: Writing your Application

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1 Relational Schema

```
Developer(did,
          dname NOT NULL,
          demail NOT NULL,
          dphone,
          PRIMARY KEY (did))
Publisher(pid,
          pname NOT NULL,
          pemail NOT NULL,
          pphone,
          PRIMARY KEY (pid))
Game(gid,
     date NOT NULL,
     rating,
     price NOT NULL,
     gname NOT NULL,
     PRIMARY KEY (gid))
Category(name,
         PRIMARY KEY (cname))
Users(uid,
      uemail NOT NULL,
      password NOT NULL,
      PRIMARY KEY (uid))
BillingInfo(uid,
            address,
            uname NOT NULL,
            payMethod NOT NULL,
            PRIMARY KEY (uid),
            FOREIGN KEY (uid) REFERENCES Users)
```

```

Transactions(tid,
            date NOT NULL,
            amount NOT NULL,
            status NOT NULL,
            uid NOT NULL,
            PRIMARY KEY (tid),
            FOREIGN KEY (uid) REFERENCES Users)

Discount(gid, startDate,
        endDate NOT NULL,
        percentOff NOT NULL,
        PRIMARY KEY (gid, startDate),
        FOREIGN KEY (gid) REFERENCES Game)

MakePublish(gid, did NOT NULL, pid NOT NULL,
            PRIMARY KEY (gid),
            FOREIGN KEY (gid) REFERENCES Game,
            FOREIGN KEY (did) REFERENCES Developer,
            FOREIGN KEY (pid) REFERENCES Publisher)

Wish(uid, gid,
    PRIMARY KEY (uid, gid),
    FOREIGN KEY (uid) REFERENCES Users,
    FOREIGN KEY (gid) REFERENCES Game)

Own(uid, gid,
    PRIMARY KEY (uid, gid),
    FOREIGN KEY (uid) REFERENCES Users,
    FOREIGN KEY (gid) REFERENCES Game)

InCart(uid, gid,
    PRIMARY KEY (uid, gid),
    FOREIGN KEY (uid) REFERENCES Users,
    FOREIGN KEY (gid) REFERENCES Game)

Friend(uid1, uid2,
    status NOT NULL,
    PRIMARY KEY (uid1, uid2),
    FOREIGN KEY (uid1) REFERENCES Users,
    FOREIGN KEY (uid2) REFERENCES Users)

BelongsTo(gid, categoryName,
    PRIMARY KEY (gid, categoryName),
    FOREIGN KEY (gid) REFERENCES Game,
    FOREIGN KEY (categoryName) REFERENCES Category)

Buy(uid, gid, tid,
    PRIMARY KEY (uid, gid, tid),
    FOREIGN KEY (uid) REFERENCES Users,
    FOREIGN KEY (gid) REFERENCES Game,
    FOREIGN KEY (tid) REFERENCES Transactions)

```

- 2 Stored Procedure
- 3 Application Program
- 4 Indexing
 - 4.1 Index 1
 - 4.2 Index 2
- 5 Visualization
 - 5.1 Vis 1
 - 5.2 Vis 2
- 6 Creativity
- 7 Teamwork